Agent Beer Game results (v5)

Final collection: 25 October 2026 · three complete non-excluded rounds per GPT model

1. **11 complete agent contexts received; 7 have passed the recorded procedure review.** Collection stopped after three complete non-excluded contexts per GPT model. No further rounds will run.

2. **H1:** Astra: lower pipeline sensitivity (-0.0002083); Sol: no mean difference; Luna: no mean difference. **H3:** Astra: higher pipeline sensitivity (+0.0002083); Sol: no mean difference; Luna: no mean difference.

3. **H2:** Astra: higher costs; Sol: lower costs; Luna: higher costs.

4. **H4a and H4b:** Education effects vary across models and contexts. Negative values mean lower order variance or bullwhip after education.

5. **Read the effect and p-value together.** The tables show signed means, reviewed context counts, raw two-sided p-values and Holm-adjusted p-values.

What each hypothesis measures

Groups: A = control reading with a separate revision opportunity; B = control reading with one committed decision; C = causal education with one committed decision. All groups share inventory information. In this AI task, fewer decision opportunities represent the proposed time constraint.

Test	Quantity in plain language	Reported comparison and meaning
H1	How much the model reduces its new order for each extra case already ordered but still on the way. This is **pipeline sensitivity**, a proxy for attending to outstanding orders.	**B−A** sensitivity. A negative difference means the constrained group reacts less to outstanding orders, the predicted underweighting pattern.
H2	The chain's **total holding and backlog cost over 24 weeks**, summed across all four roles, then averaged across chains. Holding costs 0.5 per case/week; backlog costs 1 per case/week. Units: game cost units per chain.	**B−A** cost. A positive difference means the constrained group costs more.
H3	The **same pipeline sensitivity as H1**, comparing educated and control groups while both have one decision opportunity.	**C−B** sensitivity. A positive difference means educated participants respond more to outstanding orders.
H4a	How much each role's orders fluctuate across the 24 weeks: **weekly order variance**, averaged over roles and chains. Units: cases squared.	**C−B** variance. A negative difference means education produces steadier orders.
H4b	**Weekly order variance divided by external customer-demand variance**, averaged over roles and chains. This is the dimensionless bullwhip ratio.	**C−B** ratio. A negative difference means education reduces demand amplification.

Reading the scales: A pipeline sensitivity of 1 means 24 more cases on the way produces 24 fewer cases ordered; 0 means the order does not change. The probes compare 8 versus 32 outstanding cases with other displayed facts fixed: sensitivity = (order at 8 − order at 32) / 24. This is a response measure, not an independently identified optimal order rule. A bullwhip ratio of 2 means order variance is twice customer-demand variance.

H1-H4 trends

Hypothesis	Observed trend
H1: Constraint increases underweighting	Astra: lower pipeline sensitivity (-0.0002083); Sol: no mean difference; Luna: no mean difference.
H2: Constraint increases holding + backlog costs	Astra: higher costs; Sol: lower costs; Luna: higher costs.
H3: Education reduces underweighting	Astra: higher pipeline sensitivity (+0.0002083); Sol: no mean difference; Luna: no mean difference.
H4a: Education reduces individual order variance	Astra: lower order variance; Sol: higher order variance; Luna: higher order variance.
H4b: Education reduces bullwhip ratio	Astra: lower bullwhip; Sol: higher bullwhip; Luna: higher bullwhip.

Effect sizes and p-values

Test	Model	n	Mean Δ	95% bootstrap interval	Raw p	Holm p (20)	Direction
H1	GPT-6 Astra	2	-0.0002083	[-0.0004167, 0]	1	1	Predicted direction
H1	GPT-6.1 Sol	2	0	[0, 0]	1	1	No difference
H1	GPT-6 Luna	3	0	[0, 0]	1	1	No difference
H2	GPT-6 Astra	2	38.31	[14.73, 61.88]	0.5	1	Predicted direction
H2	GPT-6.1 Sol	2	-13.82	[-27.65, 0]	1	1	Opposite direction
H2	GPT-6 Luna	3	6.433	[0, 11.85]	0.5	1	Predicted direction
H3	GPT-6 Astra	2	0.0002083	[0, 0.0004167]	1	1	Predicted direction
H3	GPT-6.1 Sol	2	0	[0, 0]	1	1	No difference
H3	GPT-6 Luna	3	0	[0, 0]	1	1	No difference
H4a	GPT-6 Astra	2	-20.44	[-29.14, -11.75]	0.5	1	Predicted direction
H4a	GPT-6.1 Sol	2	11.48	[-0.2473, 23.22]	1	1	Opposite direction
H4a	GPT-6 Luna	3	1.473	[0, 4.42]	1	1	Opposite direction
H4b	GPT-6 Astra	2	-3.165	[-4.481, -1.848]	0.5	1	Predicted direction
H4b	GPT-6.1 Sol	2	1.774	[-0.03711, 3.586]	1	1	Opposite direction
H4b	GPT-6 Luna	3	0.2148	[0, 0.6444]	1	1	Opposite direction

Here n is reviewed fresh contexts, not chains. Values are condition differences: H1/H2 = B−A; H3/H4 = C−B. Predicted signs are H1 negative, H2 positive, H3 positive, H4a/b negative. H1/H3 use outstanding-order sensitivity as a proxy for underweighting. A zero contrast means no treatment difference in this measurement.

P-values use the preselected two-sided exact paired sign-flip test. With three independent contexts its smallest possible p-value is 0.25. Bootstrap intervals are pointwise and descriptive in this small pilot. Collection ends at the requested three rounds regardless of the p-values.

Absolute sensitivity puts the H1/H3 differences in context. The table below averages each arm over reviewed contexts. Equal sensitivity across arms means no measured treatment difference; small differences should be read relative to these levels. These probe slopes are measures of response to outstanding orders.

Test	Model	n	Mean Δ	95% bootstrap interval	Raw p	Holm p (20)	Direction
GPT-6 Astra	1.003854	1.003646	1.003854				
GPT-6.1 Sol	1.000000	1.000000	1.000000				
GPT-6 Luna	0.000000	0.000000	0.000000				

Collection status

Model	Complete non-excluded contexts	Reviewed for inference	Complete excluded attempts
GPT-6 Astra	3/3	2	0
GPT-6.1 Sol	3/3	2	0
GPT-6 Luna	3/3	3	2

Fifty chains inside one context are dependent. Additional fresh contexts provide replication; increasing a within-context prefix does not create independent students. **This tests decision opportunities in fictional students.** It does not establish binding human clock pressure, normative underweighting, or real student behavior.

Observed H1-H4 differences

Requested model	Run	H1 Δ sens.	H2 Δ cost	H3 Δ sens.	H4a Δ var.	H4b Δ ratio	Review
GPT-6 Astra	run001	0.1425	148.2	-0.001667	-23.38	-3.502	Pending: incomplete orchestration record
GPT-6 Astra	run002	0	61.88	0	-29.14	-4.481	Reviewed
GPT-6 Astra	run003	-0.0004167	14.73	0.0004167	-11.75	-1.848	Reviewed
GPT-6.1 Sol	run001	0	36.69	0	-12.33	-1.993	Pending: incomplete orchestration record
GPT-6.1 Sol	run002	0	-27.65	0	23.22	3.586	Reviewed
GPT-6.1 Sol	run003	0	0	0	-0.2473	-0.03711	Reviewed
GPT-6 Luna	run001	0	8.36	0	0.0209	0.003169	Excluded: Public state fields omitted before decisions
GPT-6 Luna	run002	0	0	0	0.07625	0.01026	Excluded: Code generated current-page choices
GPT-6 Luna	run003	0	7.45	0	0	0	Reviewed
GPT-6 Luna	run004	0	11.85	0	4.42	0.6444	Reviewed
GPT-6 Luna	run005	0	0	0	0	0	Reviewed

Expected directions are H1 negative, H2 positive, H3 positive, and H4a/b negative. A difference in that direction alone is not significance. H1/H3 use pipeline sensitivity as a proxy.

Open these files

- Current presentation figures, PDF/SVG exports and short captions.

- Short extracted hypothesis conclusions.

- Live notebook: run all cells and tick Auto-refresh (10 s).

- Per-context / condition results and per-chain extracted results.

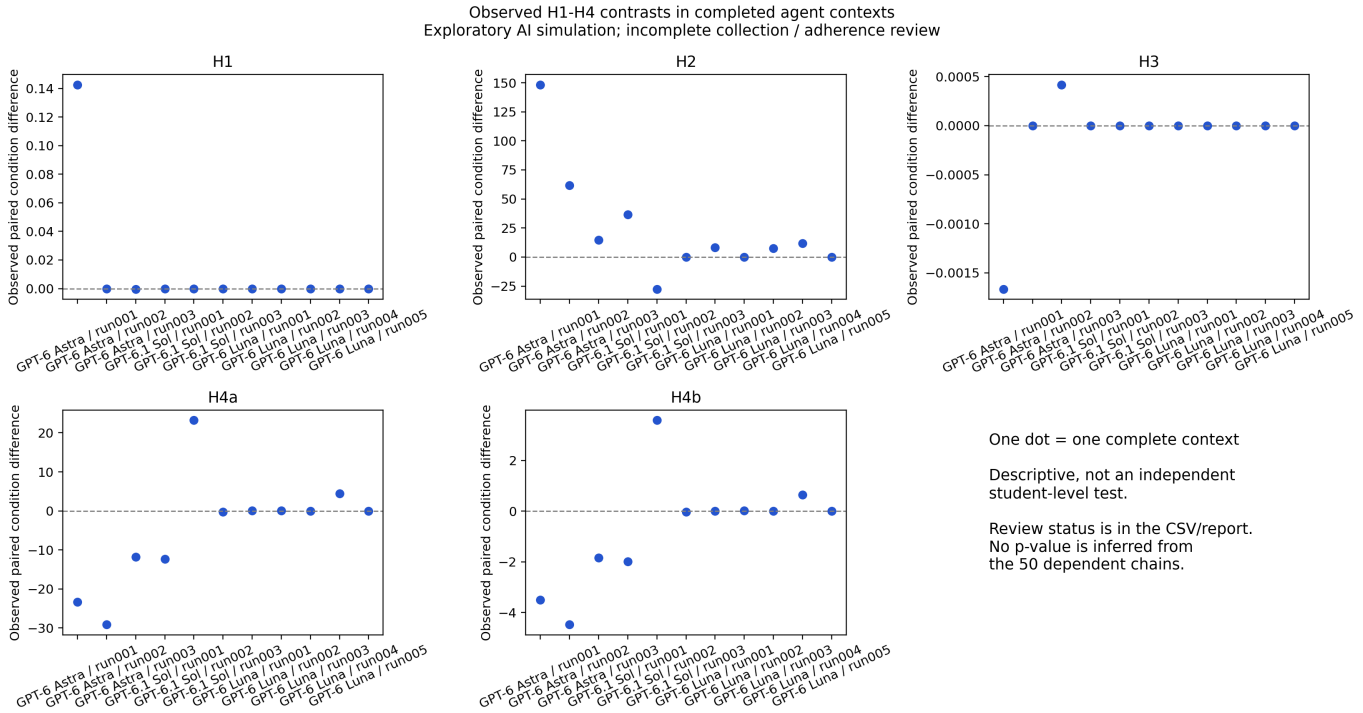
- Current reviewed-context estimates, intervals and adjusted p-values. Missing p-values mean replication or review is insufficient.

- Within-context size table: first 5/10/20/30/40/50 chains, descriptive only.

- Fresh-context size table: original attempted prefixes 1/2/5/10 plus observed time-cutoff sizes; eligible counts and missingness remain explicit.

- H1-H4 versus chain-count plot and data: observed direction stability within contexts.

- Exact tasks, prompts, decisions and public trace reviews.



Observed contrasts. Individual complete contexts; review/exclusion status is listed in the preceding tables.

Collection and limits

Requested agent models are GPT-6 Astra, GPT-6.1 Sol and GPT-6 Luna, all high effort. Client model labels/settings are recorded separately from provider authentication. These agent observations are not pooled with manual website or historical API data. Claude is still a manual route.

A has a matched control reading plus a separate revision opportunity; B has the control reading and one committed decision; C has causal education and one decision. All three arms share a context, allowing carryover. Python computes state; the model supplies literal order pages. Full games replay before extraction. A code execution log does not by itself prove prompt adherence.

Procedure review covers visible prompts and tool interactions, complete public state exposure, and actual literal decisions. Encrypted parent-message bodies are covered only where exact operator-saved plaintext receipts exist; these are operator attestations, not decrypted or independently authenticated provider transcripts. Missing receipts remain explicit review gaps. This review does not validate student fidelity or clock pressure. Deviating attempts are preserved and do not disappear from the attempted sample.

When eligible replication exists, the exploratory tests use paired context differences, two-sided sign flips and pointwise bootstrap intervals. The Holm family retains all 20 model-by-endpoint tests, including missing Claude tests. These assumptions do not make model contexts equivalent to real human samples. Repeated live looks are not a stopping rule.

No new direct API calls are made by this monitor, analysis or local transport. The shared API ledger is unchanged; agent-service usage is separate.

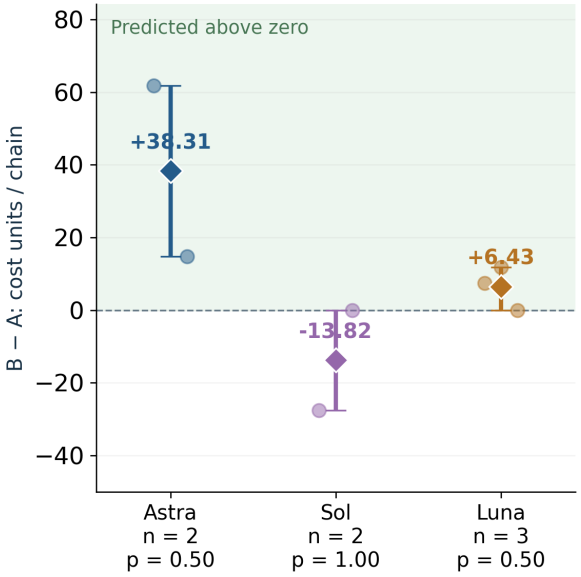
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Model comparison: reviewed results

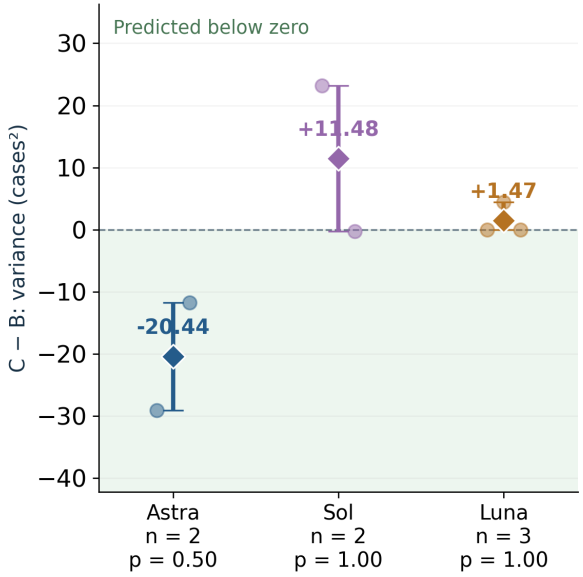
Model comparison: final reviewed results

Small dots = separate runs · Diamonds = model means · Bars = observed minimum–maximum across runs

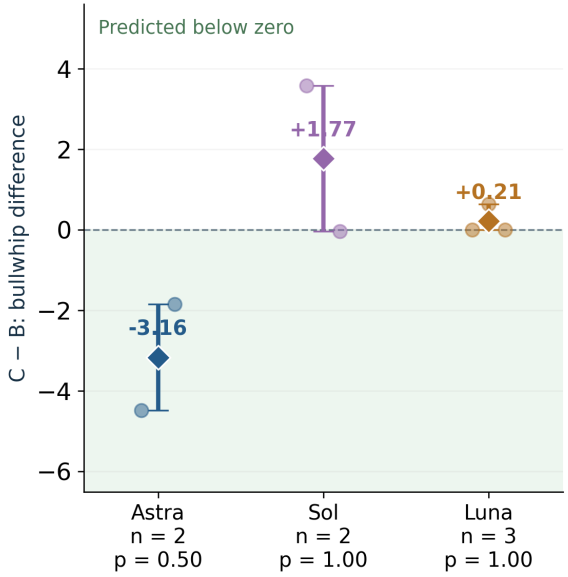
H2 Total cost



H4a Order variance



H4b Bullwhip ratio



Same reviewed runs as the statistics table: Astra 2, Sol 2, Luna 3. No lines connect independent rounds.

A = control + revision B = control + one decision C = causal education + one decision

Printed p-values are raw, two-sided exact tests; all Holm-adjusted p-values = 1.00. Bars show run-to-run range.

Small dots show individual reviewed contexts; diamonds show model means; bars show the observed minimum–maximum across reviewed runs. The printed p-values are raw two-sided exact tests. This uses the same reviewed observations as the statistics table: Astra2, Sol2, Luna3. A = control + revision; B = control + one decision; C = education + one decision.

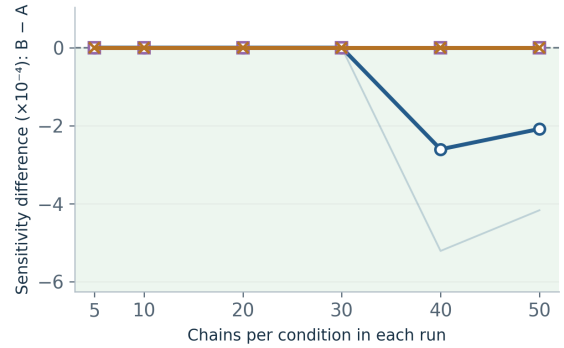
H1–H4 metrics versus chain count

How estimates change with chains per condition

Bold: reviewed model means · Faint: individual reviewed runs · At 50 chains: same means as the statistics table

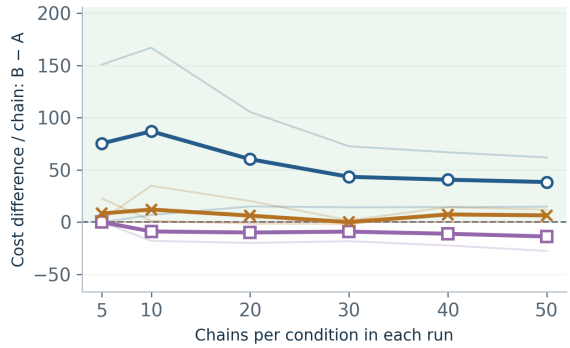
H1 Outstanding-order sensitivity

Predicted: below zero



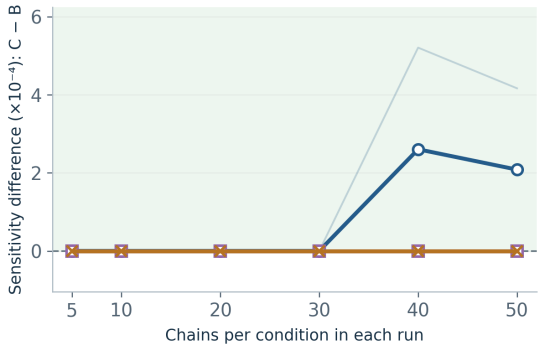
H2 Total inventory + backlog cost

Predicted: above zero



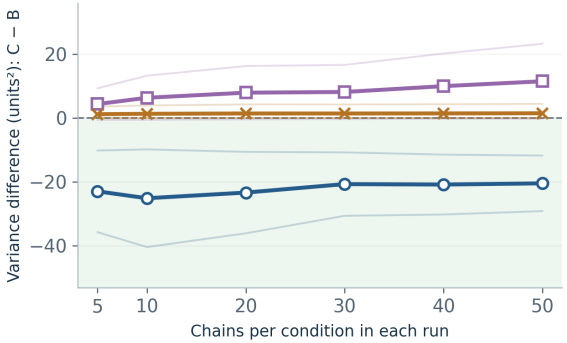
H3 Outstanding-order sensitivity

Predicted: above zero



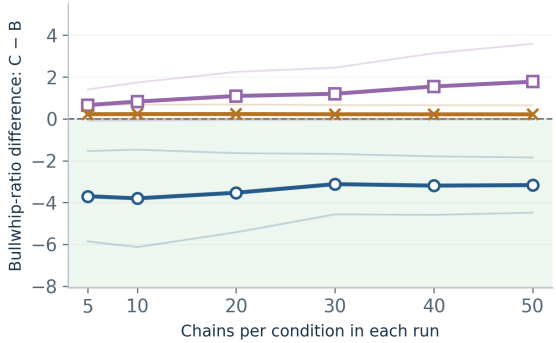
H4a Individual order variation

Predicted: below zero



H4b Bullwhip ratio

Predicted: below zero



Same sample as the statistics

- GPT-6 Astra (n = 2 contexts)
- GPT-6.1 Sol (n = 2 contexts)
- GPT-6 Luna (n = 3 contexts)

How to read the axes

X: first k chains in each run
Independent n stays at 2 / 2 / 3
Green: predicted effect direction
H1 / H3: magnified ×10⁻⁴ scale

A = control + revision B = control + one decision C = causal education + one decision

Reviewed runs only. H1/H3 differences are tiny in absolute terms. Chain prefixes describe subset stability; no extra independent rounds are added.

Bold curves average the reviewed contexts for each model; faint curves show individual reviewed runs. Each point uses the first 5–50 chains per condition in every included run. Independent n stays2/2/3. H1/H3 use a magnified scale. All 50-chain endpoints match the statistics table.